Waypoint

Your all-in-one sports tool for setting smart goals, connecting with your team, and improving at the things that matter to you.

Project Overview

PROJECT BACKGROUND

Youth Mountain Biking

Our project centered on the National Interscholastic Cycling Association (NICA), a youth mountain biking association with leagues and teams across the country. NICA values not only athletic performance, but also building strong minds, bodies, character, and community. For this project, we collaborated with the South City Composite Otters, a NICA team in St. Louis with athletes from grades 6–12.

OPPORTUNITY SPACE

Supporting Endurance Building

Youth mountain bikers are learning how to build endurance while balancing school, rest, and motivation. Because their bodies are still developing, it can be hard for them to know when to push and when to recover.

Most fitness apps are designed for adults and focus on performance and competition. These tools often ignore how rides feel mentally and physically, and they rarely support healthy recovery or confidence for younger riders.

The challenge was to design an experience that helps youth mountain bikers build endurance in a holistic way, supporting body, mind, and community, while encouraging long-term engagement with the sport.



GUIDING QUESTION

How might we design experiences that support youth mountain bikers in building endurance in ways that sustain body, mind, and community health?

Research

Interviews

INTERVIEW 1

Sarah Pointer

Our interview with Sarah Pointer, a coach for NICA's Otters team, highlighted a holistic and community-centered approach to youth mountain biking. She emphasized that physical skill development, mental resilience, and social connection are deeply connected and essential to overall well-being.

This interview first introduced us to the importance of a tight-knit team environment and a small coach-to-athlete ratio. These factors allow coaches to offer more personalized support while fostering a strong sense of belonging, which shapes how young athletes grow both on and off the bike.

KEY QUOTES

"Hanging out and talking is a bonding experience between kids and coaches."

"We are very holistic: training, eating, and recovering."

"Mental struggles are real ... Encourage the kids with positive self-talk."

"I would love an app that joins up all the tools we use ... or something a little less intense for kids who are training."

INTERVIEW 2

Mattie White, M.D.

Mattie White shared her perspective as a parent, physician, and first-time coach, showing that NICA is focused on inclusion, safety, and personal growth. Even with no prior experience, both Mattie and her daughter Nieva gained confidence through a supportive team environment.

A main takeaway was the "challenge by choice" mindset, where athletes are encouraged to try new skills but are not forced to do things they are not ready for. Small improvements help keep athletes motivated, while too much comparison or pressure can be discouraging.

Community was described as the most important part of the experience. A positive team culture and strong coach support help athletes feel comfortable, motivated, and safe while learning.

KEY QUOTES

"The kids struggle with effort sometimes, if they don't feel up to doing a hard thing. But we encourage them to explain why, and tell them they can do it."

"The community is the best part. Nieva is not there to win, but to make friends and have fun."

"It's important to help kids set specific, realistic goals at the start of the season and help them understand step goals instead of huge ambitious goals."

INTERVIEW 3

River C.

In our interview with River, she shared her experience of a high school senior who began mountain biking during COVID and found it to be an important source of community, confidence, and mental clarity. Riding helped her stay active, meet friends, and manage stress.

As she gained more experience, she began to take biking more seriously and did additional training outside of practice. Over time, however, heavy competitiveness reduced her enjoyment, especially when combined with school and injury. As she stepped back from racing, her motivation shifted toward social connection and riding for fun rather than performance.

This interview highlighted the importance of flexible goal-setting, self-paced progress, and low pressure in supporting long-term engagement with the sport.

KEY QUOTES

"My biggest motivator to continue was the social aspect."

"Mountain biking has helped me learn my own body better, and learn what I need."

"When I was at peak competitiveness, I went to practice three times a week and trained outside of practice three to four times a week."

"It would be helpful to have pre-practice or pre-race reminders and checklists."

"As you increase intensity for your goal, having training targets would be valuable."

INTERVIEW 4

Gabriel E.

In this interview, Gabriel shared his experience of a senior athlete who began mountain biking during COVID and stayed involved through high school, largely due to the strong sense of community on the team.

While competition initially motivated his participation, Gabriel described a shift toward valuing balance, enjoyment, and long-term engagement. Listening to his body, managing rest and recovery, and balancing training with school and other commitments were important in sustaining motivation.

The interview highlighted the importance of flexible goal-setting and supportive coaching over constant performance pressure.

KEY QUOTES

"I try to listen to my body. I like to train pretty hard when I can, but I know when I need a rest, am not feeling well, or have to focus on another commitment."

"It varies by athlete—some do it as a hobby, some competitively, and some as recreation, so it's a middle ground to be learned by everyone."

"A couple years ago, I wasn't doing so well that season ... I had to learn to accept the outcome if I did my best."

"You get to know the coaches all pretty well ... It's such a big, connected community ... Some coaches have better experience with training."

Contextual Observation

Team Practice

We attended a team practice to better understand how the youth athletes learn skills in a real training environment. During this visit, we interviewed Dennis, a parent and coach on the South City Otters, who shared insight into how skills are taught through repetition, coach demonstration, and supportive feedback. We observed that athletes often stop to talk through difficult sections of the trail before trying them multiple times, which helps build confidence and improvement quickly.

The practice environment was intentionally low-tech and community-focused, with athletes staying present and engaged with one another. This enforced the importance of keeping digital tools off the bike and using them instead to support reflection, goal-setting, and preparation beyond practice.

KEY OBSERVATIONS & QUOTES

Shout-out circle at the end of practices

There is a structured skills curriculum

"Head coaches talk to kids to get a sense of their goals ... it's hard to keep track of everyone's goals if there are a lot of kids."

"The kids are motivated for a variety of reasons ... The high schoolers are more competitive and have to do training outside of practice, but the younger kids not as much."

Secondary Research

Competitive Analysis

Many interviewees mentioned that some athletes on the team use apps like Strava and Garmin to track their biking sessions and workouts. These apps provide detailed metrics such as pace, calories, and heart race, but may feel intimidating because of their high intensity and granularity. While they are helpful for competitive athletes who want in-depth analysis of their performance, they are not as applicable to youth athletes who are looking for a more lightweight, fun way to track their progress.

The team also uses TeamSnap to coordinate, but many interviewees expressed frustration toward the app, citing its cluttered UX design, notification overload, and slow performance.

As we ideated and designed our app, we aimed to create something uniquely applicable to youth sports teams. Our goal was to cater toward athletes of varying skill and intensity levels while creating a seamless experience for both coaches and athletes.

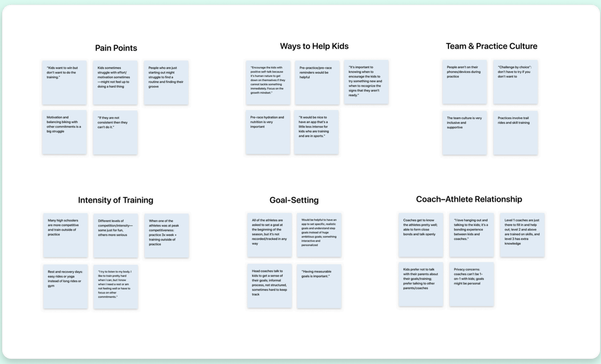
COMPETITOR PRODUCTS

GARMIN™

STRAVA

teamSNAP

Synthesis

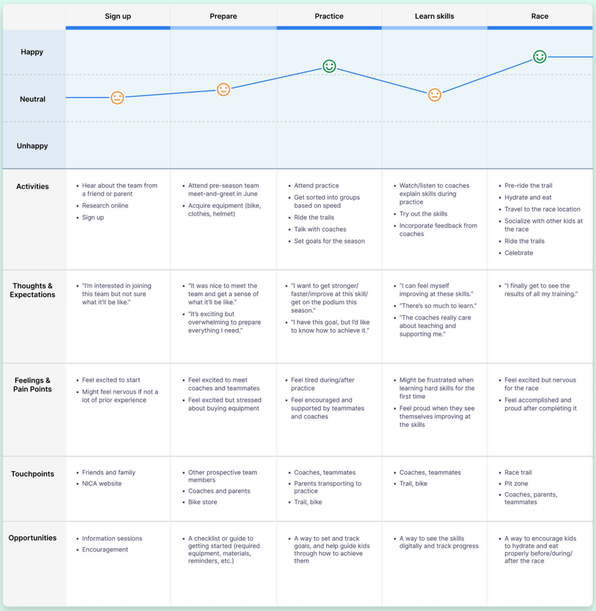


Affinity Map

Affinity mapping is a method used to organize research data, such as interview feedback and observations, into related groups. It helps reveal patterns and shared themes, allowing us to better understand users' needs and experiences.

Through this process, we discovered that coaches play an important role in creating an environment where athletes can learn and develop at their own pace. We also found that youth riders have a wide range of goals, from race performance to personal improvement or just enjoying the team community.

These insights guided our solution. Rather than focusing on a single definition of success, we designed an experience that allows athletes to set and track personal goals at their own pace. By breaking progress into small, manageable steps, the solution supports different motivations while encouraging sustainable development.



Current-State User Journey Map

The current state user journey shows how youth mountain bikers experience training and skill development today, and where challenges often arise. It helps us understand moments of frustration as well as opportunities for improvement.

Through this process, we realized that learning new skills is often frustrating, especially when athletes cannot tangibly visualize their progress.

They may also set ambitious goals at the beginning of the season but feel critical of themselves when they do not meet those goals quickly, or they may lack the structured guidance they need to make step-by-step progress.

This reflection shaped our solution. By focusing on making skill learning feel more encouraging and rewarding, we identified a key point in the athlete's journey where a thoughtful design could have the greatest impact.



Hannah Wilson

Student mountain biker

About

14 · Ninth grade · St. Louis · Student

Fun-loving · Energetic · Curious · Caring · Creative · Determined

"I want to get more serious about mountain biking and meet my goals while still having fun."

Bio

Hannah is an ninth grade student who joined the NICA Otters as a sixth grader. Before joining, she had ridden a regular bike but did not have any mountain biking experience. Now that she's in high school, she wants to be more serious about biking. She joined the team together with a friend from school but has made new friends on the team as well. Her dad became a level 1 coach on the team after she joined.

Goals

- Get on the podium at a race this season
- Get stronger and faster
- Have fun biking with her friends

Needs

- Guidance and support from coaches to help her improve
- Motivation to keep working on her goals

Pain Points

- Struggle to stay motivated to work on her goals when she doesn't see immediate progress
- Scared of crashing and getting injured
- Training and racing is tiring and physically strenuous
- Want to win but not motivated to train
- Critical of herself when she doesn't meet her own expectations

Behaviors & Context

- She attends practice three times a week but does not currently train outside of practice.
- She travels to participate in races during race weekends.

Metrics of Success

- Confidence, enjoyment, and sense of accomplishment while biking
- Achieving her goal of getting on the podium this season
- Improved biking skills and speed

Persona

A persona helps us imagine how a real young athlete might interact with the solution based on their goals, needs, and pain points. It allows us to evaluate whether our design is truly addressing the user's problems rather than making assumptions.

As we moved into prototyping, the persona helped guide our decisions by highlighting which features would be most meaningful and useful for the athlete. This made sure that our design stayed focused on real user needs instead of unnecessary complexity.

REFINED GUIDING QUESTION

How might we help youth mountain bikers set personal goals, stay motivated by tracking and celebrating progress, and give coaches an easy way to support that progress over time?

Concept

PRODUCT CONCEPT

Based on our research, we decided to create a mobile app that guides athletes in setting smart goals, tracking their progress, and staying motivated. At the same time, the app improves team communication by allowing coaches to see athletes' goals and facilitating team-wide shoutouts and announcements.

	Onboarding	Goal-setting	Training	Progression	Results
Activities	• Download the app • Input information • Set a season goal	• App suggests smaller step-by-step goals • User selects app-suggested goals and/or adds their own • Optionally, share goals with coaches	• User follows the app's suggested plan based on their goals • Log activities and track progress • Coaches review goals, add suggestions and encouragements	• App prompts the user to assess their progress toward their goal • Move on to the next goal, or stick with the current one	• Go through the goals one by one • Review results at the end of the season
Thoughts & Feelings	• "I want to achieve this major goal but am not sure how to get there." 🤔	• "These smaller goals help me understand the steps I can take toward my big goal." • "Sharing my goals with my coaches will help them help me." 😎	• "Now, coaches can know what I would like to work on during practice just by checking the app." • "It's helpful to have clear guidance on what to do every day." 💪	• "It's really motivating to see my progress and check off goals one by one." 😄	• "My training paid off!" • "I'm so proud that I was able to achieve my goal." OR "Even though I didn't meet my big goal, I still made progress on all these smaller goals." 🥳

Future-State User Journey Map

The future-state user journey builds on the current-state journey by mapping how a young athlete feels while using the goal-setting app created in response to our "How might we" question. It shows how the experience changes when the solution is introduced.

Using our persona, we walked through the user flow step by step to ensure the experience remained mostly positive and supportive. Through this process, we discovered that completing small goals is highly motivating for young athletes, as it helps them clearly see that their training is paying off and that their larger goal is becoming closer and more achievable.

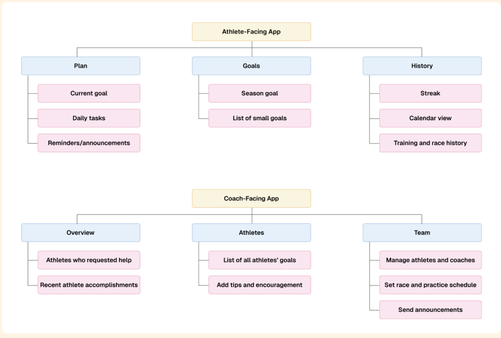


Storyboard

The storyboard shows a situation where our design solves a real problem for the persona we created. It helps explain why an athlete would choose to use the app and what kind of support they are looking for.

Through this process, we realized that the app is most effective when it supports self-motivation rather than constant supervision. The storyboard highlighted the importance of allowing athletes to work at their own pace, without feeling pressured or closely monitored.

This insight shaped our solution by reinforcing the idea that the app should act as a supportive resource, giving athletes the freedom to focus on what they want to work on, when they want to work on it.



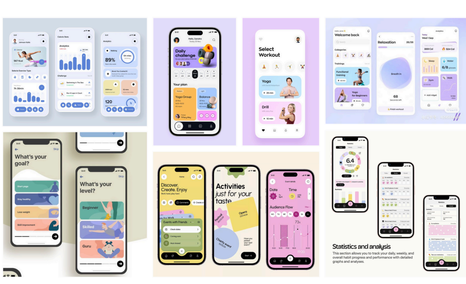
Sitemap

A sitemap is a structured overview of an app's pages that acts as a roadmap for how the experience is organized. It helped us understand the flow of the app, ensure information is easy to find, and identify any missing or unnecessary pages.

Creating the sitemap before beginning the design phase helped us organize the information hierarchy, understand how the features would relate to each other, and plan out the navigation.

Because our app features a double interface—one for athletes and one for coaches—it was especially important to determine which features would be available for which users and how the data would translate across interfaces. The information architecture differed for the two user groups based on their needs.

Although our information architecture evolved as we began designing the prototype, the sitemap was a valuable starting point to help us distill the key features and craft an intuitive experience.



Moodboard

A moodboard is a collection of visual inspiration that guides the visual direction of a design. For our moodboard, we drew from a variety of relevant products, including fitness apps, data visualizations, and apps with colorful, playful designs. We were guided by the goal of creating a visual identity that would be friendly and appealing to young users, while still emphasizing clarity and usability.

Gathering a moodboard provided us with a useful reference as we moved into the design iteration stage. We returned to it throughout our design process, drawing visual inspiration for UI elements like colors and layouts.

Design Process

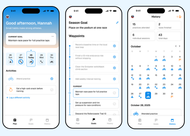
Initial Wireframes



Low-Fi Wireframes

We began the design phase with low-fidelity wireframes, which were quick sketches that helped us get a sense of how information and features might be organized on each page. We focused mostly on the athlete-facing interface, but sketched out visualizaitons of the coach side as well.

Even as we moved into the higher-fidelity stage, we continued sketching to visualize new ideas or iterate on existing layouts. These sketches allowed us to test out a lot of possibilities rapidly without getting stuck on an idea from the beginning.

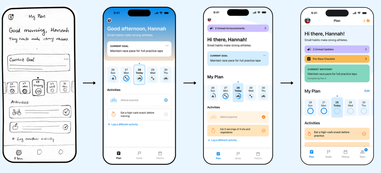


High-Fi Wireframes

We evolved the low-fidelity sketches into high-fidelity wireframes, starting with the main three pages (plan, goals, and history). We began with the set on the left, but the visuals felt too formal for a youth-centered app.

For the next iteration on the right, we incorporated a friendlier color palette and gave the cards colorful fills instead of gray outlines.

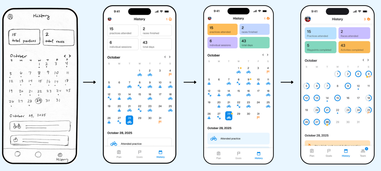
Design Evolution Highlights



Home/Plan Page

The home page went through several iterations. We knew from the beginning that this was where the athletes' daily plan would live and where they would be able to see their current goals and progress, but we adjusted the layout to make it clearer and more inviting.

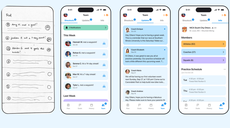
After talking with users, we learned that pre-race reminders and announcements are important for athletes to have easy access to, so we added these to the top of the home page. We also added an edit button to the plan after learning that athletes value flexibility as well.



History Page

Our app also includes a history page, which gives athletes a quick overview of their activities over time. We evolved how we represented each day on the calendar: first we used icons, but these were too visually busy and potentially confusing if a day had more than one activity.

We then shifted to using a progress ring to represent each day, to be consistent with the plan page. A dot represents a day where the user did not do the suggested plan but logged a different activity. The star and flag icons represent waypoint completion and race days.

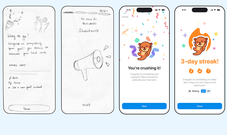


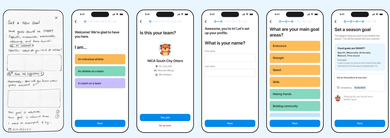
Team Page

Because several interviewees emphasized the community aspect of the team, we added a team page that includes shoutouts, updates/announcements, and crucial team info. Inspired by the shout-out circles at the end of practices, users can send "power claps" to their teammates who have accomplished a goal or streak. This system celebrates athletes for working toward their personal goals at their own pace, even if they are not winning races.

Encouragement

We wanted to make sure our app gave users motivation and positive reinforcement for making progress. That's why we added a daily streak feature to motivate them to stay consistent, as well as moments of delight such as confetti and a happy otter mascot when they complete a waypoint. Waypoints are also represented by stars, which appeals to youth, who respond well to badges and feel motivated seeing "special days."



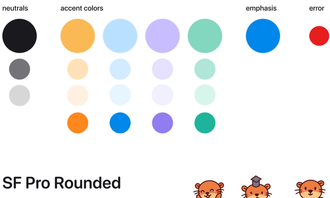


Onboarding & Goal-Setting

The onboarding flow was one of our main focuses because it was important to give users the right level of flexibility in how they wanted to use the app, and also personalize their experience based on their goals and preferred intensity level. We learned from interviews that there are both competitive and casual athletes with different kinds of goals; this is why we gave users the option to create a daily plan or not, and included goal categories not only for training but also for social, mental, and lifestyle.

The onboarding guides users through setting up their profile, setting their season goal and first waypoint, and customizing their personalized plan. At first, we grouped many steps onto a single page, but later broke them up into separate pages with a progress bar so that the process feels approachable. We also used the goal-setting interaction as an opportunity to teach users to set SMART goals through a friendly otter mascot who provides guidance.

Design System



SF Pro Rounded

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Aliquam imperdiet dictum nisl, at tincidunt urna.



We created a design system that was friendly, approachable, and clear. The color palette is vibrant and varied, with colors that appeal to youth without being childish. We also selected a rounded typeface that feels youth-friendly but clean and readable. The UI elements such as icons, cards, and buttons are all rounded as well.

We imagined that the app would allow teams to select an avatar based on their mascot. Having a friendly mascot makes the app experience more fun and delightful for young users, while also boosting team pride. The mascot appears in key moments such as setting new goals and celebrating goal completion.

Reflection

Failures, Pivots, and Takeaways

Throughout the design process, we encountered moments where our initial ideas did not fully support our users and needed to be reconsidered. One major pivot involved how goals and waypoints were created. At first, we planned to have coaches set waypoints for athletes, but we realized this would require extra work from coaches and could feel forced or controlling for young riders.

Through research and iteration, we learned that athletes are more motivated when they set their own goals. Allowing youth riders to define their own waypoints not only increased motivation, but also helped them learn how to set realistic and effective goals for themselves. This shift better aligned the solution with our focus on long-term development.

We also adjusted the visual design of the app during this phase. Early versions felt too cold and mature, so we redesigned the interface to be more friendly and encouraging. This change helped the app feel more approachable and reduced pressure, reinforcing its role as a supportive tool for youth rather than a performance tracker.

In addition, we evolved the information architecture of our app: we added a team page to the athlete interface, and broke up the three tabs of the team page into separate pages on the coach interface. As we prototyped, we realized it would be important for athletes to be able to see team information as well, and that coaches might need even easier access to this information.

Feedback Sessions

We participated in several in-class feedback sessions that played an important role in our design iteration process. In one session, we spoke with two coaches from the Soutch City Otters team and two product design experts, which gave us valuable perspectives from both domains.

One key takeaway was the idea of matching athletes with coaches based on shared interests, such as skill development or social goals, to create a more supportive and personalized community. This inspired us to add coach and athlete profile pages that display their goals and expertise, and to recommend coaches to athletes based on their goals.

We also learned how important it is to clearly communicate the design process itself, as it helps others understand how we arrived at our final solution.

In addition, we had several informal conversations and feedback sesions in class. While speaking with our instrcutor, who is also a coach on a team, we learned that athletes are split up into squads based on skill level for each practice session, and this information is distributed through “practice plans.” However, the current method of assembling and distributing these practice plans is cumbersome. This is why we gave athletes and coaches easy access to a list of squads in the app.

Constraints and Scope

Although our app concept encompasses both an athlete-facing interface and a complementary coach-facing interface, because of scope and timeline constraints, we chose to focus on the athlete-facing interface.

This allowed us to explore deeper design decisions like motivation, progress visualization, and flexible activity logging. The coach side is conceptualized in two screens to illustrate how data would flow both directions.

Roles and Responsibilities

Lasya Maganti

My role on this team was to organize and communicate. I was in charge of setting up and conducting interviews and coordinating with my team members to create a schedule for us to meet. I also assisted with analyzing our research in order to create our solution and drew the storyboard.

Natalie Rodriguez

My role on this team was to help transcribe interviews and engage in fieldwork. I also assisted in researching and comparing existing apps to better understand how our app had increased productivity and engagement in our Waypoint design.

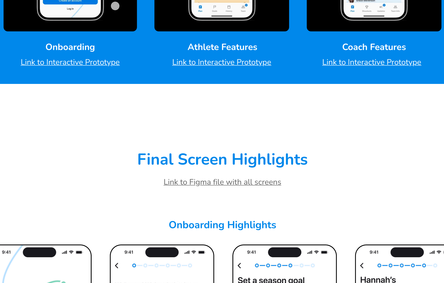
Sophie Tao

My role on this team was to synthesize the research and design the prototype. I assisted with research by transcribing interviews and conducting contextual observation. I also created the affinity map, persona, user journey maps, sitemap, and moodboard. Finally, I wireframed the mobile app, iterated on the design, and prototyped the final solution.

Final Solution

Video Walkthroughs

We recommend watching the walkthroughs to see all of the features in action!



Onboarding
Link to Interactive Prototype

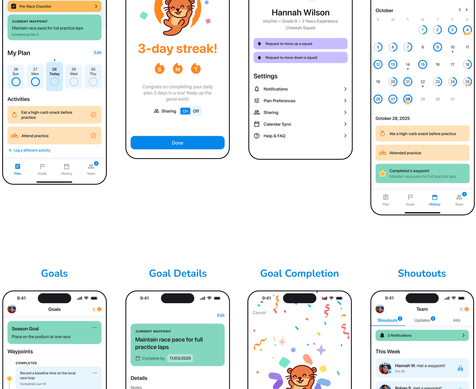
Athlete Features
Link to Interactive Prototype

Coach Features
Link to Interactive Prototype

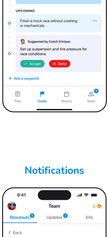
Final Screen Highlights

Link to Figma file with all screens

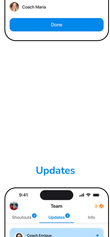
Onboarding Highlights



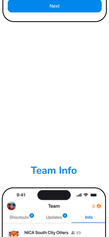
Athlete Plan



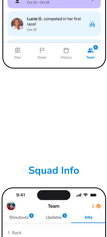
Streak



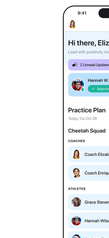
Account Info



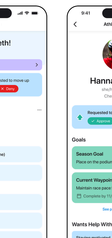
History



Goals



Goal Details



Goal Completion



Shoutouts



Notifications

Updates

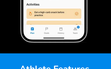
Team Info

Squad Info

Coach View: Plan



Coach View: Athlete



Coach Profile

